

REVIEW ARTICLE

A REVIEW ON PRESENT STATUS OF CHEMICAL OR ONCOGENIC TRANSFORMATION AND TUMOR STUDY METHODOLOGIES

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ABSTRACT

The possible causative interactions responsible for mutagenesis are discussed, with reference to chemical induction or oncogenic transformation. The chemical or carcinogenic induction upto microsomal level is prior stage of genotoxicity, inducers are of two types, (1) initiators and (2) promoters, initiators are involved in inheritable changes of genetic material. There are number of pathological biochemical and microbial methods useful for development, and estimation of tumor. The different methodologies to study the tumor are reviewed.

INTRODUCTION

Genotoxic carcinogens are those chemicals that exerting the demonstrable mutagenic activity, and are capable of producing radicals that bind various constituents, including DNA (Douglas *et al.*, 1988,¹ Clayson, D.B. 1989,² Perera 1984,³ Grasso and Sharratt 1991).⁴ Interaction with DNA is of primary importance, since this may lead to mutagenic events. The majority of carcinogens are electrophilic reactants or must be converted metabolically a chemically reactive electrophil that then binds covalently to cellular DNA and macro molecules (Troll and Wiesner 1985).⁵ Initiators are involved in a heritable modification of genetic material of a cell. Initiators are mutagenic activators. The examples of genotoxic carcinogens (initiators) are nitrosamine (afotoxins), mycotoxins and aromatic amines.

Number of pathological changes appeared after the implementation of initiators, e.g. cell necrosis and hyperopic degeneration, whereas subtle changes are increase in cellular organelles, in enzymatic activity or organ enlargement (Kato *et al.*, 1970,⁶ Schulte 1974).⁷

Oncogenic transformation

The oncogenic transformation of a cell is that by either DNA or RNA tumor virus. The genome of the oncogenic viruses becomes covalently linked to the cellular DNA in the transformed cells, (Croce-1977).⁸ Their transformation of cells by either DNA or RNA tumor virus is associated with the integration of either the viral or proviral DNA sequences into the cellular DNA. The transforming genes of retroviruses (A family of RNA viruses) are homologous to specific cellular genes, could induce oncogenic transformation. RNA viruses contain a reverse transcriptase enzyme that permits translation of the RNA, message of the tumor virus into the DNA code of the infected cell (Salmon and Sartorelli, 1986).⁹ A specific human T. cell leukemia virus (HTLV-1) has been identified as causative of this specific type of leukemia. A closely related virus (HTLV-III/LAV/ARV) cause AIDS (Acquired Immunity Deficiency Syndrome).

Induction

The number of chemicals induces the malignant growth of tissue that culminate cancer. The sequential administration of one or more dose of an initiator (nitrosamines) followed by long-term continuous exposure to a nongenotoxic carcinogen or promoter e.g. phenobarbital causes the modulation of hepatocarcinogenesis. The chlordecine and α -hexachlorocyclohexane increase the activity of cytochrome. P-450 and MFO system (Grasso and Hinton, 1990)¹⁰ and produce liver tumors and increase in endoplasmic reticulum (SER) clofibrate, ciprofibrate, fenofibrate and nafenopin are examples of peroxisome inducers, peroxisomes contain a range of enzyme, involved in metabolism of fatty acids (Cohen and Grasso, 1981).¹¹ Lipofuscin or age pigment appears in the lysosome within two or three weeks of treatment with fenofibrate or DEPH, this pigment is associated peroxidative (H_2O_2) activity, which is regarded as mild toxic effects (Price *et al.*, 1986,¹² Mitchel *et al.*, 1985).¹³

Tumor Induction and Inoculation of Cell Suspension

Genotoxic initiators are generally used to induce the tumor, are aflatoxin (nitrosamine), (Bannasch, 1982¹⁴ and Moor *et al.*, 1982).¹⁵ Tumor could also be produced by inoculation of cell suspension of well established cell lines on nude mice generally subcutaneously (Woerdenbag *et al.*, 1987).¹⁶ The cell suspension is produced in Hank's solution and calibrated (number of viable cells) by trypan blue exclusion or by detergent trypsin method by help of flow cytometers. Many types of cell lines are used for the purpose of evolution of new anticancer drug. Lewis lung tumor is originated in 1951 as a carcinoma of lung in C57 BL/6 mouse, its biological features are similar to some human solid tumors, like wide resistance chemotherapeutics and metastasis formation, this carcinoma is very transplantable, malignant and poorly dif-

ferentiable epidermoid. Sarcoma, fibrosarcoma, melanoma and leukemia cells are used for the same purpose of inoculation in *in vivo* test animals (Wang *et al.*, 1993,¹⁷ Lelecreeq *et al.*, 1993,¹⁸ Yamada *et al.*, 1990¹⁹ and Arisawa *et al.*, 1990).²⁰ Table 2 is given the type of cell lines, and suitable mice for inoculation.

The estimations and in vitro tests

The different criteria are used to express the growth of tumor, and therapeutic or chemopreventive parameters. Yield of cancer, is the number of malignant tumors in animal that develop cancer (Pitot and Serica 1980).²¹ Incidence, is the percentage of treated animals contracting cancer. Modulation, is the kinetics of formation of cancer, recording the incidence and yield of histologically characterized cancers. Latency of cancer development, is time period for the first cancer to appear in treated animals (Robenfred *et al.*, 1991).²² Doubling time (Td), is the calibration of progress in tumor, or the time necessary to reach the volume twice as big as the starting point. Measurement of tumor growth is the formula to calculate the content of an ellipsoid $V = 1/6\pi \times \text{length} \times \text{thickness} \times \text{width}$. The three diameters of the tumor (in mm) were obtained and converted to volume, another formula is also used, volume = length x width x height / 2 (Woerdenbag *et al.*, 1989).¹⁶ Survival time, is a relative term and expressed as percentage of mean survival time of treated mice related to that of control mice, it is generally checked on day 60, survival time = T/C (Treated mice/controlled mice), Wang *et al.*, 1993).¹⁷ Antitumor activity is the estimation of increase in life span (ILS), calculated by the formula.

$$\text{ILS} = [(T/C) - 1] \times 100\%$$

T = MSD (mean survival days of treated group)

C = MSD (mean survival days of control group)

(Yamada *et al.*, 1989).¹⁹ There are number of *in vitro* tests are used to preliminary evaluation of medicinal agents.

Tissue Culture

The minimum essential medium (MEM) augmented with 10% fetal bovine serum is used for the growth and propagation of cell lines. HEP-2, PC-13, B16, LY5187 and P-388 cells are generally used for *in vitro* tests. Cells are suspended in tissue culture plates, with test compounds dissolved in DMF (dimethylformamide) and incubated at 37°C in 5% CO₂ incubator (Arisawa *et al.*, 1990).²⁰ Recent rapid advances in tissue culture technology have made three different kinds of *in vitro* tests possible (a) drugs of unknown antitumor potential may be tested against cell derived from non human tumors (b) drugs of unknown antitumor potential may be tested against human tumor cell lines (c) drugs of known antitumor potential may be tested against human tumor samples removed at operation or biopsy. In cell culture or *in vitro* test it is clearly test to use human cells (Dawson 1973).²³

The *in vitro* mutagenic and antimutagenic activities are evaluated by salmonella/mammalian microsome test in TA98, TA100, TA1535, TA1537 and TA1538 strains of *S. Typhimurium* (Ames *et al.*, 1975).²⁴

CONCLUSIONS

Though we can enlarge many types of neoplastic disease that need not shorten life if treated with drugs and different modalities (Pantcalabresi *et al.*, 1993)²⁵ e.g. choriocarcinoma, acute leukemia, wilms tumor, Ewings sarcoma, retinoblastoma, Hodgkin disease, Mycosis fungoids, testicular carcinoma, etc. but cancer is still uncureable in majority of the cases and still required new parameters and dimensions of research. Tumor vaccine is also a comparatively new and hopeful dimension in cure of disease. The autologous tumor vaccine was prepared from freshly excised tumor cell suspension (Takita *et al.*, 1977)²⁶ this suspension was added with VCN (vibrio cholera neuramindase), the combined therapy of this and BCG vaccination give the yield of median survival of 34.7 months, which was very higher as compare 12.3 months survival time of control group received conventional management.

Table 1
Non-genotoxic inducers of different enzymes

Enzymes	Inducers	Reference
Cytochrome p.450 hydroxy tolune	Phenobarbitone, Polychlorinated biphenols, butylated	Grasso and Hinton, 1990 ¹⁰
MFO system	Phenobarbitone, Polychlorinated biphenols, butylated hydroxy tolune	Grasso and Hinton, 1990 ¹⁰
Peroxisome inducers	Clofibrate, nafenopin fenofibrate	Cohan and Grasso 1981 ¹¹
Age pigment lipofuscin (H ₂ O ₂)	Lipofuscin, DEPH	Price <i>et al.</i> , 1986 ¹² Mitchel <i>et al.</i> , 1985

Table 2
Tumor cell lines used for the purpose of inoculation

Tumor cell lines	Suitable Media for inoculation	General Mode of inoculation	Reference
Lewis lung carcinoma	C57BL Mice	Cell suspension	Woerdenbag <i>et al.</i> , 1987 ¹⁶
Sarcoma-180 tumor	ICR/SLC Mice	Cell suspension	Wang <i>et al.</i> , 1993 ¹⁷
Ehrlich tumor cells (ELT-Line)	C57BL/6J Mice	Cell suspension	Leclercq <i>et al.</i> , 1993 ¹⁸
Meth A. fibrosarcoma IMC carcinoma MM-46 carcinoma	ICR Mice CDF C31 and Mice	Cell suspension	Yamada <i>et al.</i> , 1990 ¹⁹
Human lung carcinoma (PC-13) Mouse melanoma B16 Leukemia cells L5178Y Mouse P388 leukemia cells Human larynx carcinoma HEP-2	B ₆ D2F, Mice	Cell suspension	Arisawa <i>et al.</i> , 1990 ²⁰

REFERENCES

1. Douglas G.R.L., Blakey D.H., Clayson D.B. (1988). *Mutat. Res.* 96: 83.
2. Clayson D.B. (1989). *Mutat Res.* 221: 53.
3. Perera F.P. (1984). *Environ. Res.* 34: 175.
4. Grasso P. and Sharratt M. (1991). *Annu. Rev. Pharmacol. Toxicol.* 31: 253.
5. Troll W. and Wisner R. (1985). *Ann. Rev. Pharmacol. Toxicol.* 25: 509.
6. Kato R., Takanaka A., Onoda K.I. (1970). *Jpn. J. Pharmacol.* 20: 572.
7. Schulte-Hermann R. (1974). *CRC Crit. Rev. Toxicol.* 3: 97.
8. Croce C.M. (1977). Identification of oncogenic chromosomes in transformed human cells. In neoplastic transformation: Mechanism and consequences, ed. H. Koprowski, pp.69-85. Berlin: Dahlenkon frenzen.
9. Salmon S.E. and Sartorelli A.C. (1986). Cancer chemotherapy in basic and chemi-

- cal pharmacology, ed. B.G. Katzung. p.665, California, Norwall (Conm los Attos 1986).
10. Grasso P. and Hinton R. (1990). *Mutat. Res.* (In press).
 11. Cohen A.J., Grasso P. (1981). *Food Cosmet. Toxicol.* 19: 585.
 12. Price S.C., Hinton R.H., Mitchel, F.E., Hall D.E., Grasso P. (1986). Time and dose study on respons. of rats to the hypo.
 13. Mitchell F.E., Price S.C., Hinton R.H., Grosso P., Bridge J.W.B. (1985). *Toxicol. Appl. Pharmacol.* 81: 371.
 14. Bannasch P., Moor M.A., Klimek F., Zeban H. (1982). *Toxicol. Pathol.* 10: 19.
 15. Moore M.A., Mayer D., Bannasch P. (1982). *Carcinogenesis* 3: 1429.
 16. Woerdenbag H.J., Lemistra W., Hendriks H., Malingre M., Th., Konings A.W.T. (1987). *Planta. Med.* 318.
 17. Wang J.Z., Tsumura H., Ma N., Shimura K., Ito H. (1993). *Planta. Med.* 59: 54.
 18. Leclercq J.Q., Bouzahzah B., Pons A., Gerimers R., Angenot L., Bassleer R., Barbason H. (1993). *Planta. Med.* 59: 59.
 19. Yamada H., Komiyama K., Kivohara H., Chol-Cyong J., Hirakawa Y., Otsuka Y. (1990). *Planta. Med.* 56: 182.
 20. Arisawa M., Fujita A., Naokata M., Saburo K. (1990). *Planta. Med.* 56: 377.
 21. Pitot H.C., Serica A.E. (1980). *Int. Rev. Cytol.* 101: 125.
 22. Roberfroid M.B., Delzenne N., Preat V. (1991). *Toxicol.* 31:163.
 23. Dawson M. (1973). *Pharm. J.* (April 4): 315.